

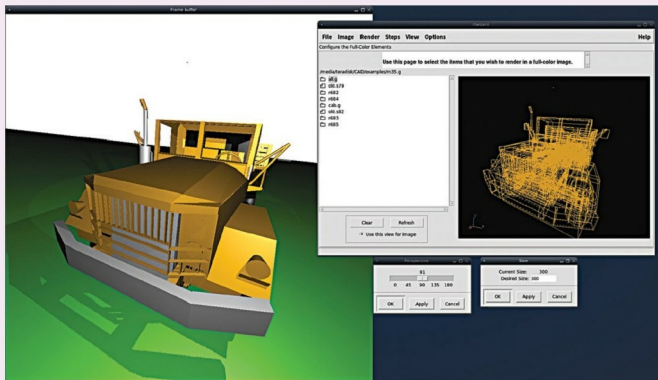


Fig. 2: Mesh network for a 3D figure (Image courtesy: www.abclinuxu.uz)



Fig. 3: Ray tracing support for graphics rendering and geometric analysis (Image courtesy: www.cs.utah.edu)

be used for pure animation where speed is the requirement.

Thus, it is capable of optical effects like reflection, refraction, scattering or dispersion required in 3D modelling for proper visualisation.

Scripting is easily possible for writing programs in special run-time environments that can be operated manually one by one. BRL-CAD uses Doxygen source code to change or move the commands on screen.

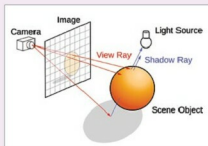


Fig. 4: Optical effects for 3D modelling (Image courtesy: Wikipedia)

View angle in BRL-CAD. The viewing angle in BRL-CAD is set as per parameters like azimuth, elevation, twist and amount of zoom. And the image capture can be carried out even when the solid shape is rotating at 360-degree angle.

The shape is viewed in MGED and saved by BASH script.

The BRL-CAD screen can be skipped directly, but the user needs to follow MGED command window to come out of the tool completely. MGED is the geometry editor.

Image processing and signal processing tools. BRL-

CAD provides tools to visualise signals and images in 3D modelling system.

BRL-CAD 7.26 version

NURBS. NURBS provides interoperability with other CAD systems, allowing reduced model sizes.

GCV and ICV. These plugin architecture tools for geometry and image conversion are intended to replace all BRL-CAD conversion and process functionality with an easy universal interface for users and application developers.

FASTGEN and DATUM. The new FASTGEN exporter implements extensive compatibility feature mapping for round-trip geometry conversion. Datum objects provide new modelling support for reference points.

System requirements

BRL-CAD 3D modelling and solid work tool requires a system with multi-core processor. Else, the system might hang or slow down due to heavy files. **EFY**